

Invited Paper: Solving Fine-Grained Static 3DIC Thermal with ML Thermal Solver Enhanced with Decay Curve Characterization

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ABSTRACT

Static chip thermal analysis provides detailed and accurate thermal profile on chip. The chip power map, commonly modeled as rectangular regions of distinct heat sources, significantly impacts the chip thermal profile. Since the heat sources result from numerous cells in functional blocks, the design space of chip power map is prohibitively enormous. Numerical simulations can be reliable for solving complex power maps; however, it could be very time-consuming when simulating a large SoC and/or 3DIC designs. Thus, there is an urgent need for speeding up the static chip thermal analysis to tackle various power maps. In this paper, we propose an approach of integrating our developed machine learning thermal solver [1] and decay curve characterization for solving static chip thermal with diverse power maps. The machine learning thermal solver would first solve the power maps on a coarse level (e.g., 200 μm). The thermal results are further enhanced using the decay curve algorithm which would fine tune the solution locally provided by the machine learning thermal solver and calculate the local temperature variations at a finer level (e.g., 10 μm). The deep learning models are trained on augmented artificial power maps and tested on realistic chip power maps. Experimental results validate the effectiveness of the proposed approach of offering fast and accurate chip thermal profile.

CCS CONCEPTS

• Artificial intelligence • Machine learning • Modeling and simulation

KEYWORDS

Chip thermal, Machine Learning, Decay curve, ML thermal solver

1. Introduction

Recent advances in technologies such as wafer level front-end 3DIC chip stacking, for example, TSMC's SoIC (System on Integrated Chips), Chip-on-Wafer (CoW) or Wafer-on-Wafer (WoW), stackings can also be assembled with backend 3DIC technologies such as Chip on Wafer on Substrate (CoWoS) or Integrated Fan-out (InFO) as shown in Figure 1 [2]. Other foundries offer similar technology roadmaps [3].

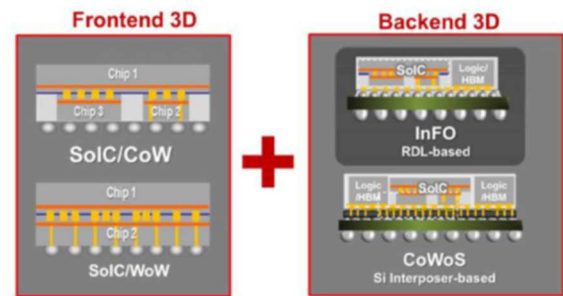


Figure 1 Industry trends for holistic 3D Heterogenous Integration

Several emerging design optimization techniques on thermal considerations are needed for the new generation of 3DIC with closely packed chiplets and strong cross-die thermal coupling. The most important is the use of Dynamic Thermal Management (DTM) with thermal throttling via DVFS (Dynamic Voltage Frequency Scaling) techniques. Traditionally, very few on-chip thermal sensors are placed and usually not on optimal locations for SoC designs due to the use of coarse mesh grid on-chip in chip-package-system static/transient thermal simulation [4]. However, advanced 7nm/5nm designs such as the Vega 2.0 3DIC from AMD have 64 on-chip thermal sensors in place as shown in Figure 2. This implies that an accurate static and transient thermal simulation is needed with fine mesh grid to optimize the placement of on-chip thermal sensors.

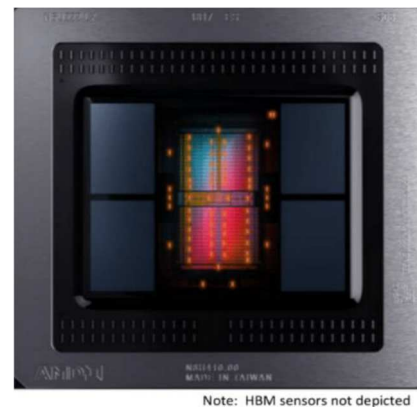


Figure 2 Large number of on-chip thermal sensors are placed in advanced CPU/GPU 3DIC such as the Vega2.0 from AMD for DTM through fine-grain DVFS control

It is well understood that thermal issues can severely degrade the performance and reliability of a chip and thermal runaway may occur if the temperature-dependent leakage power increases exponentially with increasing local temperature at certain areas in a 3DIC. A large value of peak temperature and thermal gradient on-chip caused by localized hotspot and cross-die thermal coupling can have a severely negative impact on transistor performance, stress, aging, electromigration (EM), voltage drops, and timing. A few of the unique challenges and use cases for thermal modeling and simulation are outlined below.

- a. Fine-grained static and transient thermal analysis on large 3DIC designs is required and demands adaptive meshing or machine-learning technology to overcome the limitation using traditional CFD/FEA based solvers.
- b. Heterogeneous integration 3DICs may consist of analog/mixed-signal and digital designs which have very different thermal and stress requirements that need to be co-optimized among chiplets and packages in 3DIC.
- c. Testing of large 3DICs consisting of CPU/GPUs, etc. presents a major challenge due to multiple localized thermal hotspots and dynamic voltage drop affecting yield. Co-optimization of test techniques and localized thermal hotspots and Vdroop on 3DIC should be considered[19][20].
- d. For Silicon Photonics 3DICs, accurate thermal gradient analysis is required for the co-optimization of the 3DIC package and the required thermal heater for PIC design.

Although the above is not an exhaustive list of use cases, it clearly indicates a need for accurate thermal modeling and simulation for 3DIC. For these thermal challenges in 3DICs one of the critical gaps is a methodology for a fast fine-grained thermal analysis which is a key enabling component of the above use cases. Recently, Deep Learning [17] based techniques are being explored for developing efficient thermal analysis solutions [1][6][7][16][18].

This work demonstrates a novel ML thermal solver combined with decay curve characterization for fast fine grained thermal analysis. The unique contributions of this paper are as follows:

1. A complete flow for extremely fast (order of $\sim 100\times$ runtime improvement over FEA based methods), fine-grained static thermal analysis of 3DICs.
2. A new method to account for not only same die horizontal thermal coupling, but also vertical thermal coupling among stacked dies.
3. The vertical thermal coupling can be accounted for any arbitrary stacked die depth, i.e., across any number of dies.
4. A novel auto-encoder based ML thermal solver for generating a relatively coarse-grained thermal profile for any die in the 3DIC stack.

5. Decay curve characterization and its subsequent use for local thermal coupling on the same die and for vertical thermal coupling. This generates a local fine grained thermal profile and accounts for local heating effects.
6. A framework for combining the ML thermal solver coarse grained results with the decay curve based local fine grained thermal profile to generate a complete fine-grained thermal profile.

2. Chip Thermal Analysis

Chip thermal analysis may involve multiple components of the chip and various system parameters with high dimensionality and huge parameter space. This study focuses on chip models based on simplified three-layer chip designs consisting of a Si substrate, an insulation layer, and an interconnection layer. A schematic of two stack dies is illustrated in Figure 3.

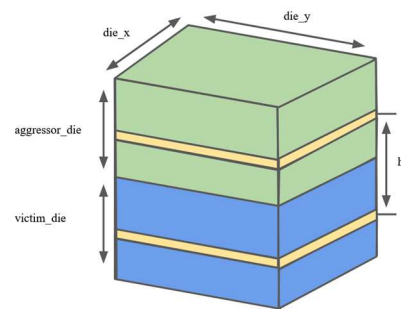


Figure 3 Schematic of two stack dies by considering vertical coupling. Each die is modelled using 3-layer model. The h represents the distance of the power layer location between aggressor die and victim die.

An arbitrary tile-based power map, which consists of rectangular regions of different heat sources, each representing a different functional block on the chip, is sandwiched between the insulation layer and interconnection layer. For instance, a $4000\text{ um} \times 4000\text{ um}$ chip $200\text{ um} \times 200\text{ um}$ tile size will lead to a 20×20 array power map. An example of such a power map is shown in Figure 4. The temperature distribution of the chip, especially the heating plane where the power map is applied, is of significant interest. In practice, such tile-based power could be random and incur drastic power gradient, which makes it challenging for conventional data-driven based approaches to learn, not to mention generalization to any unseen random power maps. Additionally, the temperature distribution on the heating plane is greatly affected by system parameters such as HTCs distribution on the boundary, the effect of which is difficult to capture. Even though uniform HTC boundary condition is commonly adopted in research and industrial applications for simplification, this assumption would not help capture the detailed thermal behavior in chip package systems. Therefore, in this paper, more challenging and practical use cases, in which the HTCs specified on the top and bottom boundaries also have a spatial tile-based distribution similar to that of the power map. In this way, the cooling effect of air is modeled using HTC distributions on top and bottom surfaces.

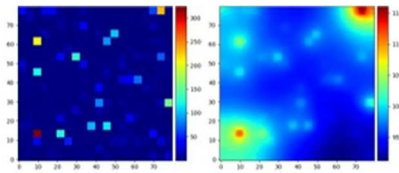


Figure 4 Example of power map and temperature on chip.

3. Flow Overview

This work develops a multiple steps process to generate fast thermal profiles on a 3DIC for all the dies in the stack. Figure 5 shows the main steps involved in generating a fine-grained thermal profile given a 3DIC and power profiles. The various parts of this flow are described below.

Inputs: The inputs required for this flow are the 3DIC configuration including the package and PCB and various material properties.

Boundary Conditions: The boundary conditions for each die in the stack is generated by performing a very coarse grained FEA thermal solve of the complete 3DIC to generate the boundary conditions for each die in the stack.

3-layer die model: With the given inputs the flow constructs a simplified 3-layer model for each die in the stack. This is the 3-layer model described in the previous section.

ML Thermal Solver: The ML thermal solver is an autoencoder based trained Deep Neural Network (DNN) model which takes the power maps and boundary conditions as inputs for any die and generates a coarse-grained thermal profile, for example at a granularity of 200umx200um [1].

Decay-curve generation: Decay curve captures how temperature changes with distance from a heat source on a given die. This decay curve characterization can be done as a pre-characterization step or a trained DNN model can be used to generate the curve.

Power Maps: The power maps contain an array of heat sources arranged as tiles on a fine-grained grid. Typically, each tile is of the size 10umx10um.

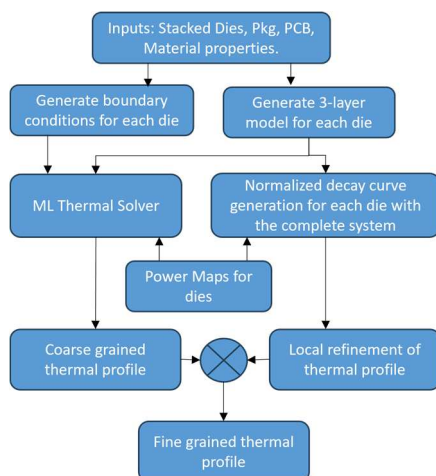


Figure 5 Main steps in generating a fine-grained 3DIC thermal profile

Coarse-grained thermal profile: The boundary condition, the 3-layer model and the power map of a die are the inputs to the ML thermal solver. Using these inputs, the ML thermal solver generates the coarse-grained thermal profile of the die. Typically, a 200umx200um resolution is used for the coarse-grained thermal profile.

Local refinement of the thermal-profile: The decay curve is used for calculating the local heating effects for a more accurate fine-grained thermal effect. This step calculates only the local effect, typically at a 10umx10um resolution from near-field neighbors.

Fine-grained thermal profile: The fine-grained thermal profile is generated by combining the coarse-grained thermal profile and the local heating effects calculated from the local refinement step.

The details of each of the above steps are provided in the following sections.

4. Machine Learning Thermal Solver

The detailed description for the ML thermal solver components is elaborated in [1]. The general workflow consists of four major components:

a). Decomposition of computational domain

The computational domain for the 2D chip power map is decomposed into fixed-size patches. As prescribed in [1][5], based on an extensive ablation study, patch size of 64x64 is used for the 1024x1024 computational domain, which is also adopted in this investigation. Since this study focuses on chip power maps as well as distributed HTCs, decomposition is applied on the computational domain for all input and output variables, power map, top HTC, bottom HTC, and temperature field.

b). Encoding on patches using an Auto Encoder

The patches obtained from the decomposition are combined and encoded by pre-trained encoders. These encoded latent vectors are concatenated based on the patches' adjacency information and would be used for the next step.

c) Constrained ML Thermal solver: A constrained equilibrium problem is solved using fixed point iterations where the encodings on each patch and its neighbors are evaluated iteratively using a pretrained flux conservation autoencoder. The fixed-point iteration is stopped when a certain condition is met, like conventional numerical solvers.

d) Decoding on patches: After above mentioned fixed-point iteration is converged, the latent vector for temperature would be collected and decoded into temperature field. Once the temperature fields are retrieved for all the patches, the temperature solution for the entire computational domain would be established based on the previously created patch adjacency information.

5. ML Thermal Solver Enhanced with Decay Curve Characterization

The concept of decay curve can be understood by considering a single point heat source on the chip and the resulting steady-state temperatures at varying distance from the heat source. As illustrated by the black curve in the example of Figure 6, the temperature peaks at the heat source location (distance: 0 um, reference temperature: 100 °C),

then drops drastically near the heat source, and finally plateaus at the locations far away (far field) from the heat source. A zoomed in view of the decay curve (red curve in 6, 0-100um) reveals that the curve manifests as a half bell-shaped curve surrounding the heat source. It is assumed that the decay behavior is the same along different directions with respect to the heat source. Based on the characteristics of the decay curve, it could be utilized as an indicator of the influence of the heat source on its surrounding region (near field and far field). Furthermore, a similar concept has been employed in a DNN-based fast static on-chip thermal solver [6] to capture near-field coupling and the developed solver can achieve runtime reduction of greater than 100-1000x with good accuracy compared to the traditional FEA based method.

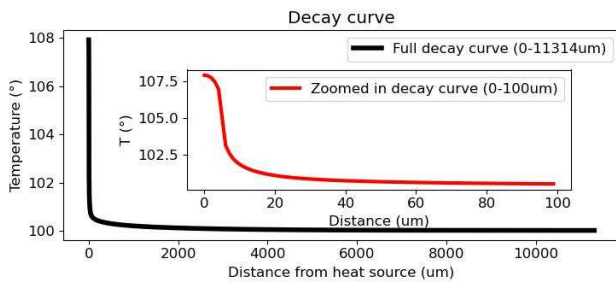


Figure 6 Example decay curve with a single heat source

5.1 Decay Curve characterization

In this work a commercial FEA based 3DIC thermal analysis software [21] is used to calculate the decay curve for the 3-layer die model. The 3-layer die model for each individual die is derived from the geometry properties, material characteristics, and boundary conditions. The decay curve characterization is done by thermal simulation of a heat source positioned on a 10umx10um tile at the center of the victim die. The temperature probe locations are placed every 1um starting from the heat source in the x and y directions. This captures the thermal effect of a heat source at various distances on the chip. The resultant decay curve is normalized with respect to unit power inputs. In scenarios involving stacked dies, separate decay curves are computed for the victim die and the aggressor dies. This approach captures the thermal interactions between different layers of the semiconductor structure.

5.2 DeepONet based decay curve prediction

Inspired by the universal approximation theorem [8][9][10], deep operator networks (DeepONets) [11] was proposed to learn nonlinear operators accurately and efficiently from relatively small datasets. DeepONets consists of a branch net and a trunk net. The branch net is used to encode the input functions and the trunk net functions to encode the locations for the output functions.

Since the decay curve is affected by various parameters modelling the chip, such as chip dimensions, input power magnitude, material properties etc., the branch net of the DeepONets could serve as a surrogate function (input function) parametrized by the chip parameters. On the other hand, as demonstrated in Figure 6,

the decay curve is also determined by the distances between the heat source and locations on the chip, which can be encoded by the trunk net. Moreover, it has been validated in [8] that DeepONets exhibits exceptional performance on small datasets. Therefore, DeepONets was selected for learning the relationship between the chip parameters as well as distance information and the corresponding decay curve.

Based on the description above, a DeepONets based neural network for decay curve prediction has been established and the network structure is depicted in Figure 7. The branch net will take system parameters as input and the trunk net will take distances as input. The information from branch net and trunk net are combined through a merging layer and output the corresponding temperature/decay curve.

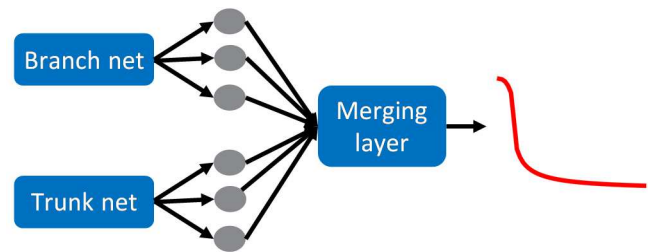


Figure 7 Schematic of the built DeepONet

5.3 Decay curve based DelaT calculation

The decay curve is used to capture the local heating effect on the victim die at all the locations and the local vertical coupling effect on the victim die from the neighboring dies. The final temperature is determined by taking into account the combined impact of both local field changes and far-field effects (ML thermal solver). The calculation of the local field temperature is carried out with a resolution of 10umx10um. The local temperature change (i.e. DeltaT) of a victim tile is computed while considering its adjacent tiles (e.g., 21x21 adjacent tiles) at the same resolution.

$$dist_{victim} = \sqrt{(dx^2 + dy^2)} \quad (1)$$

$$coeff_{victim} = f(dist_{victim}) \quad (2)$$

$$local_T = \sum_{tile_i} coeff \cdot Power_{victim} \quad (3)$$

In this scenario, the variables dx and dy in Equation (1) denote the distances from the analyzed power tile in the x and y directions, respectively. The function f is the normalized decay curve in Equation (2) and Equation (3) summarizes all the thermal impact from neighboring tiles.

Coefficients for this computation are derived from the normalized decay curve. Conversely, the far-field influence is calculated with a resolution of 200umx200um from the ML thermal solver.

In scenarios involving stacked dies, the vertical thermal coupling effects come into play. The final temperature change is also due to thermal coupling from neighboring aggressor dies. Assuming there is no dielectric material or air gap between the victim die and its neighbors, the vertical and horizontal thermal behaviors are considered identical. Conversely, when an air gap or dielectric material exists between the victim die and its neighbors, the coupling effect can be characterized using FEA analysis. This work assumes that the vertical thermal decay behavior is identical

to the horizontal one, hence the decay curve computed in the horizontal direction can be used to compute the vertical thermal coupling too.

While computing the temperature change due to local heating effects, the analysis takes into account the 21x21 power tiles of neighboring dies. It is worth noting that this local field computation is composed of two parts, viz., temperature changes caused by the victim die itself, and the other originating from neighboring dies due to vertical thermal coupling. The coefficients are calculated separately for each neighboring die and aggregated by considering the power tiles of each aggressor die in the 21x21 tile array centered at the victim tile location.

$$dist_{agg} = \sqrt{(dx_{agg}^2 + dy_{agg}^2 + dz_{agg}^2)} \quad (4)$$

$$coeff_{agg} = f(dist_{agg}) \quad (5)$$

$$local_T = \sum_{tile_i_victim} coeff_{victim} Power_{victim} + \sum_{tile_i_aggressor} coeff_{aggressor} Power_{aggressor} \quad (6)$$

In this context, dx_{agg} , dy_{agg} , dz_{agg} denote the distances between the power tiles of the aggressor die and the center of the tile under consideration within the victim die in Equation (4).

This local heating effect is computed at 10umx10um resolution. This local temperature change of a victim tile temperature is computed from the neighboring 21*21 tiles at the same resolution. The coefficients are computed from the normalized decay curve.

The total temperature change at any given tile on any die is the linear superposition of the coarse grained (200umx200um) thermal results and the local heating effects as computed above. It should be noted that the $Power_{aggressor}$ and $Power_{victim}$ are appropriately adjusted power values to avoid double counting the heating effect already considered during ML thermal solver thermal profile calculations. This is done by subtracting the uniform average power density in the 200umx200um region from the local power density of the 10umx10um tiles in Equation (5)(6).

5.4 SSIM for power map evaluation

The structural similarity index measure (SSIM) [12] was proposed for quantifying the differences between a distorted image and a reference image. The formula for calculating the SSIM between two images (of equal size) x and y is given in Equation (7), where μ_x is the pixel sample mean of x and μ_y is the pixel sample mean of y ; σ_x^2 is the variance of x , σ_y^2 is the variance of y and σ_{xy} is the cross-correlation of x and y . c_1 and c_2 are two variables to stabilize the division with weak denominator, which are represented by Equation (8) and (9), where L is the dynamic range of the pixel-values, and $k_1 = 0.01$, $k_2 = 0.03$ by default.

$$SSIM(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (7)$$

$$c_1 = (k_1 L)^2 \quad (8)$$

$$c_2 = (k_2 L)^2 \quad (9)$$

Due to its computational simplicity and its relative analytic tractability, SSIM has been widely used for image and video applications [13][14]. Additionally, since the power maps in the chip thermal applications can be treated as image signals, SSIM can be utilized as the similarity measure for comparing any two power maps. For instance, given any unforeseen power map in real applications, a max SSIM score could be obtained by comparing this power map with those in the training dataset. A relatively large SSIM value indicates a high similarity between the test power map and the ones seen by the ML model at training stage, which is likely to lead to a satisfactory model prediction

and vice versa. A threshold could be identified in advance for classifying the calculated SSIM and adjusting the strategy of the thermal solver, which is illustrated in Figure 8. For example, if the obtained max SSIM for a test power map is larger than the threshold, there is higher confidence in the ML model's prediction, on the other hand, if the calculated max SSIM is smaller than the threshold, a FEA based solve would be done on the coarse mesh to replace the ML model prediction and then another decay curve model would be incurred to solve on the fine mesh.

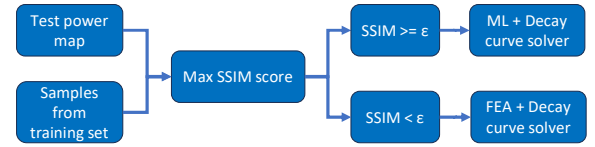


Figure 8 Thermal solver workflow based on SSIM.

6. Experiments, Results and Discussions

To validate the methodology in this work two sets of experiments were performed. The first set of experiments were performed on a single die setup to evaluate the accuracy of the methodology. The experimental setup is shown in Figure 9 and the key parameters of the setup are shown in the Table 1. Standard values of material properties were used for thermal conductivities of silicon, silicon dioxide and dielectric.

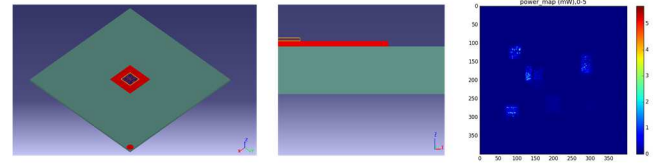


Figure 9 Top view, side view and the power map for the single die setup

The single die setup was analyzed with a power map as shown in Figure 9 using a commercial FEA based 3DIC thermal analysis software to generate a reference golden result with a 10umx10um mesh grid resolution.

Table 1 Single die test setup

Die Size	Resolution	HTC	Amb. Temp.
4mmx4mm die size with substrate thickness of 100um	10um mesh resolution	1000W/C/m ² top/bottom of die	27 C

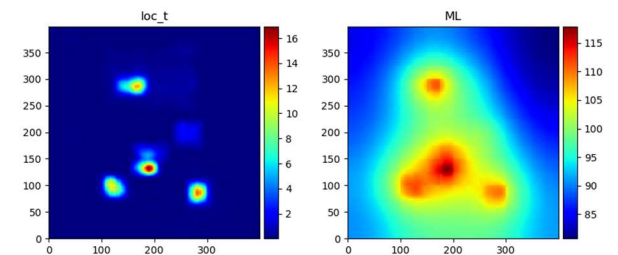


Figure 10 The local heating effect (left) and the coarse grained global heating effect (right)

The same setup was then analyzed using the proposed methodology and the thermal profile results were generated for a 10umx10um resolution. Figure 10 shows the two components of the thermal profile, viz., the decay curve based calculation of the local heating effect and the coarse grained thermal profile generated from the ML thermal solver with 200um resolution.

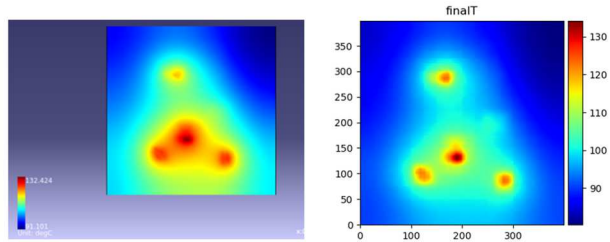


Figure 11 Comparing commercial FEA based 3DIC thermal analysis tool results (left) with the results from the proposed methodology (right) with the same temperature scale

Figure 11 compares the *golden* FEA based thermal profile with the thermal profile generated from the proposed methodology which shows an excellent agreement between the two profiles.

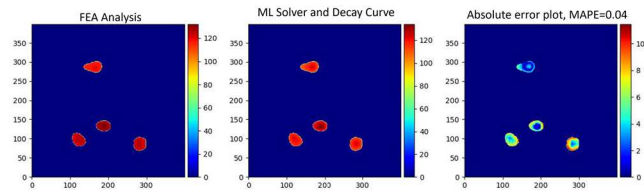


Figure 12 Error comparison between FEA based thermal analysis and ML thermal solver with decay curve based method

Figure 12 shows the 2-D error contour plot for the regions where the local heating effect is greater than 5 deg Celsius. The error plot shows mean average percentage error of 4% indicating an excellent accuracy of the proposed methodology when compared with the FEA based software tool [21].

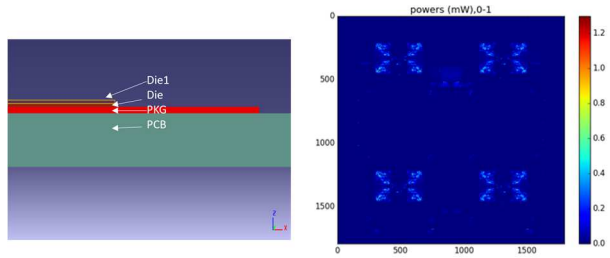


Figure 13 Setup for a stacked 2-die case and the power map on the bottom die. Top die has a uniform power distribution with a total of 1W power.

Figure 13 shows the setup for the stacked 2-die 3DIC test case, where both dies are 4mmx4mm and with 100um substrate thickness. The vertical distance between the device layer in both dies is 20um. The power distribution on the bottom die is as shown in the Figure 13 and the power distribution on the top die is assumed to be a uniformly distributed power for a total of 1W. The HTC on the die interface is calculated to be 1000W/C/m². Figure 14 shows the thermal profile results for the 2-die case on the bottom die including the deltaT from the localized heating effect from the same die, vertical thermal coupling effect on top

die and the coarse-grained ML solver which captures the global effect at the coarse-grained accuracy.

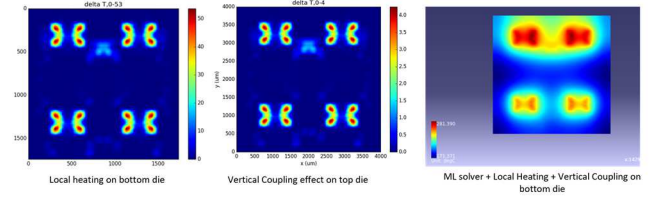


Figure 14 Thermal profile results for the 2-die case (4mmx4mm)
Combining all the three thermal profiles generates a complete fine grained thermal profile accounting for the all the effects.

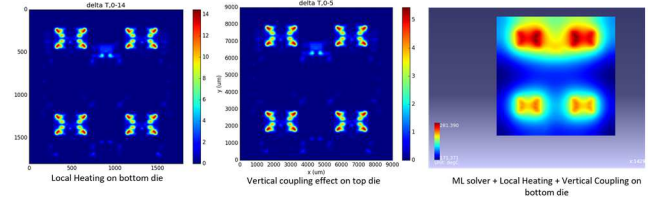


Figure 15 Thermal profile results for the 2-die case (9mmx9mm)
Figure 15 shows thermal profile results for another 2-die case but with a larger die size of 9mmx9mm for a similar power map. A key benefit of the proposed methodology provides significant runtime and capacity improvements.

Table 2 Runtime improvement using ML thermal solver enhanced with decay curve based thermal analysis

Test Case	FEA Based w/ 10um mesh	ML Solver and Decay Curve	Speed-up
Single Die 4mmx4mm	25m19s w/ 63cores 3G (mem)	1m 43s 10G (mem)	14.7X
Stacked Dies 9mmx9mm	6hr14m w/ 63cores 29G (mem)	4m 52s 10G (mem)	76.8X

Table 2 shows the runtime comparison between the FEA based thermal analysis and ML thermal solver with decay curve based thermal analysis. The runtime improvement for single die case is 14.7X and for the stacked die case is 76.8X. As the die size increases and package and PCB becomes more complex, the FEA based analysis becomes prohibitively expensive for fine grained thermal analysis. The ML solver with decay curve will provide even more speed up in such cases.

7. Conclusions

This paper presented a novel methodology by combining ML thermal solver with decay curve characterization in achieving fine grained steady state thermal analysis flow. The runtime improvements are very significant, and it is expected that this will scale with the design size. This work also described a novel method for considering vertical thermal coupling between stacked dies with decay curve in a 3DIC structure. The results were compared against a commercial FEA based 3DIC thermal analysis software indicating excellent accuracy of the methodology and faster runtime. The future work will extend this methodology to transient thermal analysis for 3DIC [22].

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